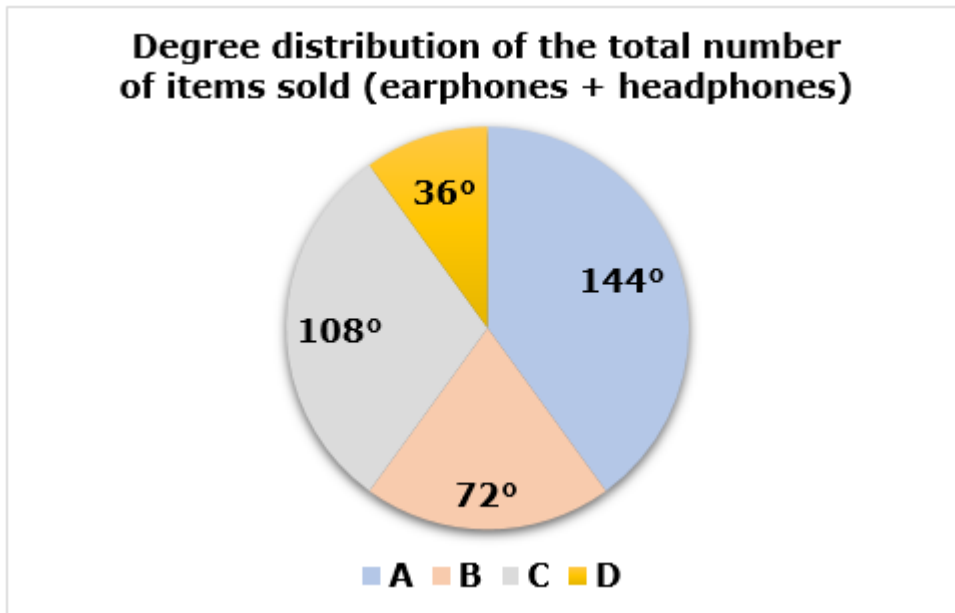


1. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of items sold (earphones + headphones) by four shops A, B, C and D. The table chart given below shows the total revenue generated (in Rs.) from the sale of all items (earphones + headphones) and the price of each earphone (in Rs.) and each headphone (in Rs.) sold by each shop.



Shop	The total revenue generated (in Rs.) (earphones + headphones)	The price of each earphone (in Rs.)	The price of each headphone (in Rs.)
A	70000	100	300
B	28000	120	160
C	44000	120	200
D	11000	70	150

Note:

@QuizByMickey

The total number of items sold = the number of earphones sold + the number of headphones sold

The total revenue generated = (the total number of items sold) * (the price of each item)

The ratio of the number of earphones sold by shop A and B is 5:3.

The number of headphones sold by shop A is 50% more than shop B.

Find the ratio of the total revenue generated from the sale of all earphones by shop A and the total revenue generated from the sale of all headphones by shop C.

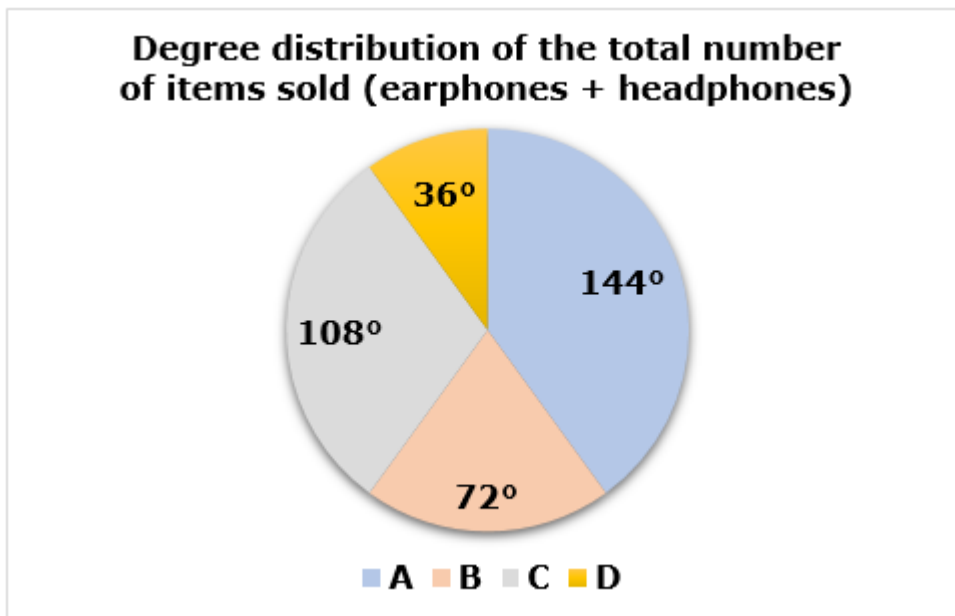
a. 3:2

- b. 2:1
- c. 4:3
- d. 5:4
- e. 1:1

2. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of items sold (earphones + headphones) by four shops A, B, C and D. The table chart given below shows the total revenue generated (in Rs.) from the sale of all items (earphones + headphones) and the price of each earphone (in Rs.) and each headphone (in Rs.) sold by each shop.



Shop	The total revenue generated (in Rs.) (earphones + headphones)	The price of each earphone (in Rs.)	The price of each headphone (in Rs.)
A	70000	100	300
B	28000	120	160
C	44000	120	200
D	11000	70	150

Note:

The total number of items sold = the number of earphones sold + the number of headphones sold

The total revenue generated = (the total number of items sold) * (the price of each item)

The ratio of the number of earphones sold by shop A and B is 5:3.

The number of headphones sold by shop A is 50% more than shop B.

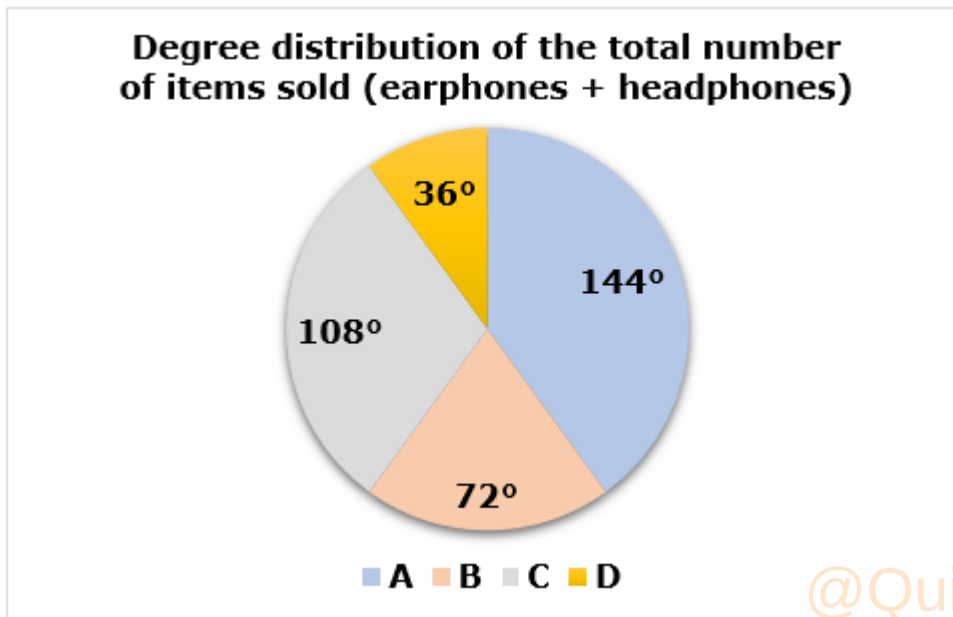
In shop B, if the ratio of the total number of earphones and headphones manufactured is 5:3 and the number of earphones unsold is 30 more than the number of headphones unsold, then find what percent (approx.) of the total number of items (earphones + headphones) manufactured by shop B are sold?

- a. 52%
- b. 30%
- c. 65%
- d. 25%
- e. 78%

3. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of items sold (earphones + headphones) by four shops A, B, C and D. The table chart given below shows the total revenue generated (in Rs.) from the sale of all items (earphones + headphones) and the price of each earphone (in Rs.) and each headphone (in Rs.) sold by each shop.



Shop	The total revenue generated (in Rs.) (earphones + headphones)	The price of each earphone (in Rs.)	The price of each headphone (in Rs.)
A	70000	100	300
B	28000	120	160
C	44000	120	200
D	11000	70	150

Note:

The total number of items sold = the number of earphones sold + the number of headphones sold

The total revenue generated = (the total number of items sold) * (the price of each item)

The ratio of the number of earphones sold by shop A and B is 5:3.

The number of headphones sold by shop A is 50% more than shop B.

In shop D, if the number of headphones sold to females is half of the number of earphones sold to females and the number of earphones sold to males is 10 less than the number of headphones sold to males, then find the number of earphones sold to males by shop D.

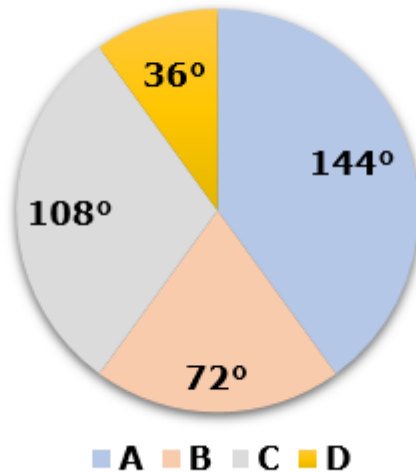
- a. 30
- b. 25
- c. 40
- d. 15
- e. 10

4. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of items sold (earphones + headphones) by four shops A, B, C and D. The table chart given below shows the total revenue generated (in Rs.) from the sale of all items (earphones + headphones) and the price of each earphone (in Rs.) and each headphone (in Rs.) sold by each shop.

Degree distribution of the total number of items sold (earphones + headphones)



Shop	The total revenue generated (in Rs.) (earphones + headphones)	The price of each earphone (in Rs.)	The price of each headphone (in Rs.)
A	70000	100	300
B	28000	120	160
C	44000	120	200
D	11000	70	150

Note:

The total number of items sold = the number of earphones sold + the number of headphones sold

The total revenue generated = (the total number of items sold) * (the price of each item)

The ratio of the number of earphones sold by shop A and B is 5:3.

The number of headphones sold by shop A is 50% more than shop B.

In shop A, the manufacturing cost of each headphone is Rs. [?] and shopkeeper suffered an overall loss of 10% on all headphones sold. $\frac{2}{5}$ th of the total number of headphones manufactured by shop A remained unsold.

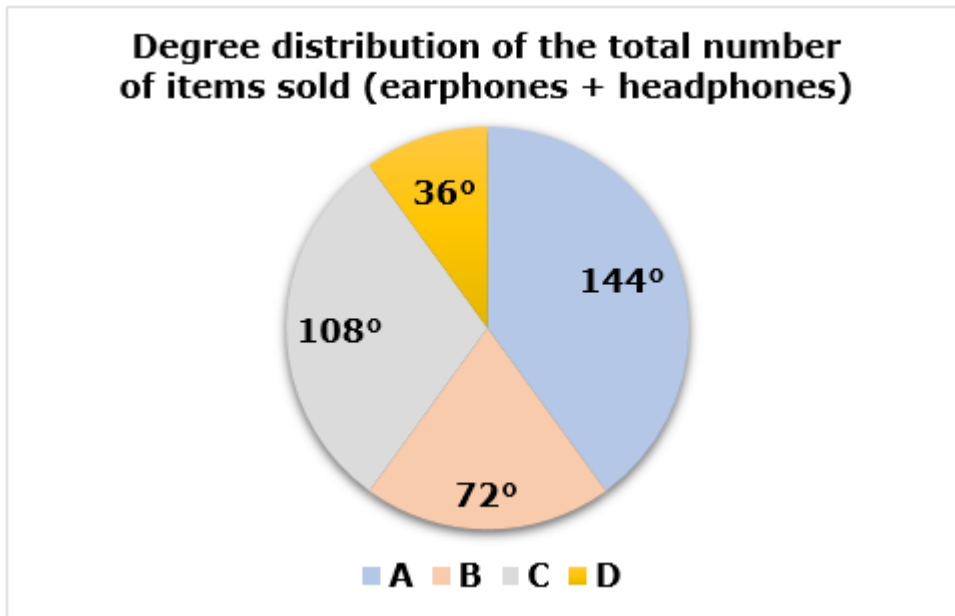
What should come in place of question mark?

- a. 150
- b. 200
- c. 100
- d. 250
- e. 400

5. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of items sold (earphones + headphones) by four shops A, B, C and D. The table chart given below shows the total revenue generated (in Rs.) from the sale of all items (earphones + headphones) and the price of each earphone (in Rs.) and each headphone (in Rs.) sold by each shop.



Shop	The total revenue generated (in Rs.) (earphones + headphones)	The price of each earphone (in Rs.)	The price of each headphone (in Rs.)
A	70000	100	300
B	28000	120	160
C	44000	120	200
D	11000	70	150

Note:

The total number of items sold = the number of earphones sold + the number of headphones sold

The total revenue generated = (the total number of items sold) * (the price of each item)

The ratio of the number of earphones sold by shop A and B is 5:3.

The number of headphones sold by shop A is 50% more than shop B.

Find the difference between the average number of earphones and average number of headphones sold by all four shops together.

a. 37.50

- b. 50
- c. 62.50
- d. 25
- e. 80

6. Questions

Read the following information carefully and answer the questions based on it.

Certain number of people speaks two different languages English and Hindi in three different towns X, Y and Z. Each person speak at least one of these languages.

Town X: The total number of people who speaks English is 120 more than the total number of people who speaks Hindi. The number of people who speaks only English is 200. The number of people who speaks both English and Hindi is 40 more than the number of people who speaks only Hindi.

Town Y: The number of people who speaks only Hindi is $\frac{3}{5}$ th of the number of people who speaks only English. The total number of people who speaks English is 350, which is 40% more than the total number of people who speaks Hindi.

Town Z: The number of people who speaks both English and Hindi is 20% less than that of town Y. The ratio of the total number of people who speaks English and the total number of people who speaks Hindi is 13:6. The number of people who speaks only English is 140 more than the number of people who speaks only Hindi.

In town Z, if the total number of females who speaks Hindi is $\frac{1}{4}$ th of the total number of females who speaks English and the total number of males who speaks English is 50 more than the total number of males who speaks Hindi, then find the total number of females who speak English in town Z.

- a. 100
- b. 120
- c. 180
- d. 160
- e. 200

@QuizByMickey

7. Questions

Find the number of people who speaks both English and Hindi in town Y is what percent of the total number of people in the same town?

- a. 40%
- b. 10%
- c. 20%

d. 50%

e. 30%

8. Questions

In another town W, the total number of people is twice of town Z, and the number of people who speak both English and Hindi in town W is 100. If the total number of people who speak Hindi in town W is 300, then find the number of people who speak only English in town W.

a. 200

b. 100

c. 400

d. 300

e. 500

9. Questions

Find the difference between the number of people who speak only Hindi in town X and the number of people who speak only English in town Z.

a. 120

b. 80

c. 65

d. 75

e. 100

10. Questions

Determine correct relation between [I] and [II].

I: The average number of people who speaks both English and Hindi in town X and Y.

II: The number of people who speaks only Hindi in town Z.

a. $I > II$

b. $I < II$

c. $I \geq II$

d. $I \leq II$

e. $I = II$

@QuizByMickey

11. Questions

The following question has two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give an answer accordingly.

Quantity I: Find the total distance covered by the bike. The time taken by the bike to cover $\frac{2}{3}^{\text{rd}}$ of the total distance with the speed of 40 km/hr is 3 hours more than the time taken by the same bike to cover the remaining distance with 25% more than its usual speed.

Quantity II: Find the total distance between point X and Y. A car started from point X at 10:00 am towards point Y, but at 12:00 pm it stopped somewhere between points X and Y. Now, at the same time a bike started from point Y towards point X and met the car at 02:00 pm. The difference between the distance covered by car and bike is 48 km, while the speed of the car is 50% more than the bike.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

12. Questions

Quantity I: Find the value of $[P^2]$. A vessel contains a mixture of 200 ml, in which the quantity of milk is $[30P]$ ml and the rest is water. If half of the initial mixture is taken out and replaced with $[P]$ ml milk and $[P + 20]$ ml water, then the ratio of the quantity of milk and water in the resultant mixture will become 8:5.

Quantity II: Find the value of $[Q^2]$. A vessel contains a mixture of $[40Q]$ ml, in which the ratio of the quantity of milk and water is 5:3. If 40 ml of initial mixture is taken out and replaced with the same quantity of water, then the quantity of milk in the resultant mixture will be $[2Q + 2]$ ml less than water.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

13. Questions

Quantity I: Find the original marked price of the article. An article is marked-up by 40% above its cost price and sold at a discount of Rs. 100. If the same article is marked-up by 20% above its cost price but the discount allowed on the article remains same, then there will be no profit no loss. (cost price remained same)

Quantity II: Find the cost price of the article. An article is sold for a profit of Rs. 40 after allowing a discount of 12%. The article was marked-up by Rs. 100 above its cost price.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II

- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

14. Questions

Quantity I: Find the value of $[M]$. The average of runs scored by a batter in first 5 matches is $[4M]$. If runs scored by him in sixth and seventh match is $[M - 3]$ and $[M - 7]$ respectively, then his average will be reduced by 25%.

Quantity II: Find the value of $[N/3]$. In a class, the average weight of $[N]$ number of students is 40 kg. If two students, whose total weight is 80 kg left the class and three new students, whose average weight is 40 kg joined the class, then the average weight of all students in the class will be $[N + 10]$. (N is positive integer)

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

15. Questions

Quantity I: Find the rate of interest offered in second year. A person invested Rs. 15000 in a scheme, which offers simple interest (per annum) for 2 years. The ratio of the rate of interest offered in first and second year is 3:2. The difference between the interest received in first and second year is Rs. 1500.

Quantity II: Find the value of $[R]$. The amount invested by a person in a scheme is Rs. 20000 at the rate of $[R]\%$ per annum compounded annually and received Rs. $[420R]$ as compound interest after 2 years.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

@QuizByMickey

16. Questions

For the following question, there is a series given, which contain one wrong term. Find the wrong term in the given series.

20, 22, 47, 144, 585, 2931

- a. 47
- b. 144

- c. 20
- d. 2931
- e. 585

17. Questions**42, 53, 39, 56, 35, 59**

- a. 42
- b. 53
- c. 39
- d. 59
- e. 35

18. Questions**28, 4, 56, 8, 112, 12**

- a. 12
- b. 56
- c. 4
- d. 8
- e. 112

19. Questions**11, 132, 300, 590, 951, 1480**

- a. 951
- b. 132
- c. 300
- d. 590
- e. 1480

@QuizByMickey

20. Questions**16, 216, 64, 100, 144, 2744**

- a. 16
- b. 144
- c. 2744

d. 64

e. 100

Explanations:

1. Questions

Let the number of earphones sold by shop A and B be $5m$ and $3m$ respectively.

Also, the ratio of the number of headphones sold by shop A and B = $150:100 = 3:2$. Let it be $3n$ and $2n$ respectively.

So, the total revenue generated from the sale of all earphones by shop A = $(5m * 100) = \text{Rs. } 500m$

And, the total revenue generated from the sale of all headphones by shop A = $(3n * 300) = \text{Rs. } 900n$

Also, the total revenue generated from the sale of all earphones by shop B = $(3m * 80) = \text{Rs. } 240m$

And, the total revenue generated from the sale of all headphones by shop B = $(2n * 160) = \text{Rs. } 320n$

Now,

$$500m + 900n = 70000$$

$$5m + 9n = 700 \dots \dots \dots (I)$$

And,

$$240m + 320n = 28000$$

$$3m + 4n = 350 \dots \dots \dots (II)$$

By solving equation (I) and (II), we get,

$$15m + 27n = 2100$$

$$15m + 20n = 1750$$

$$7n = 350$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$5m + 9n = 700$$

$$5m + (9 * 50) = 700$$

$$5m = 700 - 450$$

$$5m = 250$$

$$m = 50$$

Now, the total number of items sold (earphones + headphones) by shop A = $(5m + 3n) = (5*50 + 3*50) = (250 + 150) = 400$

@QuizByMickey

From the pie chart given above, we have,

The total value of the pie chart = 360°

Now, let the total number of items sold (earphones + headphones) by all four shops together be $360k$.

So, the total number of items sold (earphones + headphones) by shop A = 144° of $360k = 144/360 * 360k = 144k$

Now,

$$144k = 400$$

$$k = (25/9)$$

Therefore, the total number of items sold (earphones + headphones) by all four shops together = $360k = 360 * 25/9 = 1000$.

Shop A

The total number of items sold = 144° of $1000 = 144/360 * 1000 = 400$

The number of earphones sold = $5m = (5 * 50) = 250$

The number of headphones sold = $3n = (3 * 50) = 150$

The total revenue generated from the sale of all earphones = $(250 * 100) = \text{Rs. } 25000$

The total revenue generated from the sale of all headphones = $(150 * 300) = \text{Rs. } 45000$

Shop B

The total number of items sold = 72° of $1000 = 72/360 * 1000 = 200$

The number of earphones sold = $3m = (3 * 50) = 150$

The number of headphones sold = $2n = (2 * 50) = 100$

The total revenue generated from the sale of all earphones = $(150 * 80) = \text{Rs. } 12000$

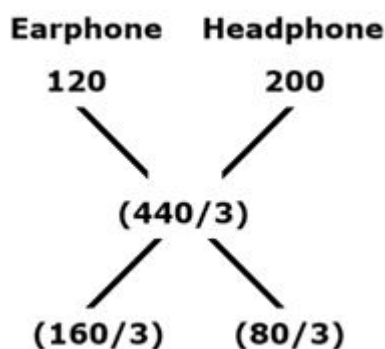
The total revenue generated from the sale of all headphones = $(100 * 160) = \text{Rs. } 16000$

Shop C

The total number of items sold = 108° of $1000 = 108/360 * 1000 = 300$

So, the average price of each item (earphones + headphones) = $(44000/300) = \text{Rs. } (440/3)$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $(160/3) : (80/3) = 160:80 = 2:1$.

The number of earphones sold = $2/3$ of $300 = 200$

The number of headphones sold = $1/3$ of $300 = 100$

So, the total revenue generated from the sale of all earphones = $(200 * 120) = \text{Rs. } 24000$

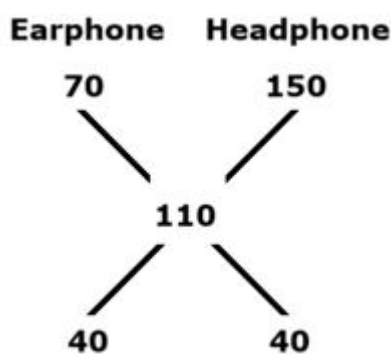
And, the total revenue generated from the sale of all headphones = $(100 * 200) = \text{Rs. } 20000$

Shop D

The total number of items sold = 36° of $1000 = 36/360 * 1000 = 100$

So, the average price of each item (earphones + headphones) = $(11000/100) = \text{Rs. } 110$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $40:40 = 1:1$.

The number of earphones sold = $1/2$ of $100 = 50$

The number of headphones sold = $1/2$ of $100 = 50$

So, the total revenue generated from the sale of all earphones = $(50 * 70) = \text{Rs. } 3500$

And, the total revenue generated from the sale of all headphones = $(50 * 150) = \text{Rs. } 7500$

Earphone

Shop	The total number of earphones sold	The total revenue generated from the sale of all earphones (in Rs.)
A	250	25000
B	100	12000
C	200	24000
D	50	3500
Total	600	64500

Headphone

Shop	The total number of headphones sold	The total revenue generated from the sale of all headphones (in Rs.)
A	150	45000
B	100	16000
C	100	20000
D	50	7500
Total	400	88500

Answer: D

Now, according to the question,

The total revenue generated from the sale of all earphones by shop A = Rs. 25000

The total revenue generated from the sale of all headphones by shop C = Rs. 20000

Therefore, the required ratio = 25000:20000 = 5:4.

Hence, the correct answer is option D.

2. Questions

Let the number of earphones sold by shop A and B be 5m and 3m respectively.

Also, the ratio of the number of headphones sold by shop A and B = 150:100 = 3:2. Let it be 3n and 2n respectively.

So, the total revenue generated from the sale of all earphones by shop A = (5m * 100) = Rs. 500m

And, the total revenue generated from the sale of all headphones by shop A = (3n * 300) = Rs. 900n

Also, the total revenue generated from the sale of all earphones by shop B = (3m * 80) = Rs. 240m

And, the total revenue generated from the sale of all headphones by shop B = (2n * 160) = Rs. 320n

Now,

$$500m + 900n = 70000$$

$$5m + 9n = 700 \dots\dots\dots(I)$$

And,

$$240m + 320n = 28000$$

$$3m + 4n = 350 \dots\dots\dots(II)$$

By solving equation (I) and (II), we get,

$$15m + 27n = 2100$$

$$15m + 20n = 1750$$

$$7n = 350$$

$$n = 50$$

@QuizByMickey

Now, put the value of n in equation (I), we get,

$$5m + 9n = 700$$

$$5m + (9 * 50) = 700$$

$$5m = 700 - 450$$

$$5m = 250$$

$$m = 50$$

Now, the total number of items sold (earphones + headphones) by shop A = $(5m + 3n) = (5*50 + 3*50) = (250 + 150) = 400$

From the pie chart given above, we have,

The total value of the pie chart = 360°

Now, let the total number of items sold (earphones + headphones) by all four shops together be $360k$.

So, the total number of items sold (earphones + headphones) by shop A = 144° of $360k = 144/360 * 360k = 144k$

Now,

$$144k = 400$$

$$k = (25/9)$$

Therefore, the total number of items sold (earphones + headphones) by all four shops together = $360k = 360 * 25/9 = 1000$.

Shop A

The total number of items sold = 144° of $1000 = 144/360 * 1000 = 400$

The number of earphones sold = $5m = (5 * 50) = 250$

The number of headphones sold = $3n = (3 * 50) = 150$

The total revenue generated from the sale of all earphones = $(250 * 100) = \text{Rs. } 25000$

The total revenue generated from the sale of all headphones = $(150 * 300) = \text{Rs. } 45000$

Shop B

The total number of items sold = 72° of $1000 = 72/360 * 1000 = 200$

The number of earphones sold = $3m = (3 * 50) = 150$

The number of headphones sold = $2n = (2 * 50) = 100$

The total revenue generated from the sale of all earphones = $(150 * 80) = \text{Rs. } 12000$

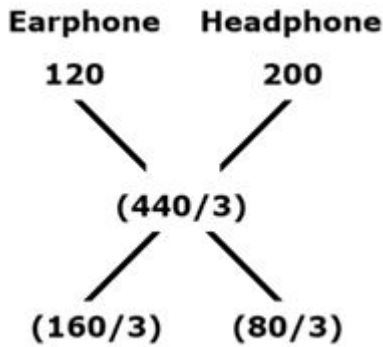
The total revenue generated from the sale of all headphones = $(100 * 160) = \text{Rs. } 16000$

Shop C

The total number of items sold = 108° of $1000 = 108/360 * 1000 = 300$

So, the average price of each item (earphones + headphones) = $(44000/300) = \text{Rs. } (440/3)$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $(160/3) : (80/3) = 160:80 = 2:1$.

The number of earphones sold = $2/3$ of $300 = 200$

The number of headphones sold = $1/3$ of $300 = 100$

So, the total revenue generated from the sale of all earphones = $(200 * 120) = \text{Rs. } 24000$

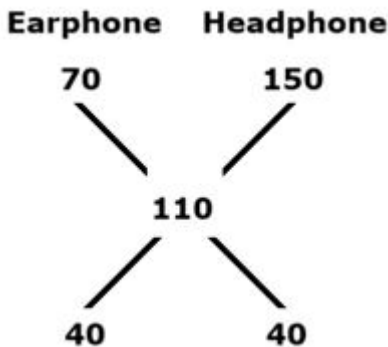
And, the total revenue generated from the sale of all headphones = $(100 * 200) = \text{Rs. } 20000$

Shop D

The total number of items sold = 36° of $1000 = 36/360 * 1000 = 100$

So, the average price of each item (earphones + headphones) = $(11000/100) = \text{Rs. } 110$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $40:40 = 1:1$.

The number of earphones sold = $1/2$ of $100 = 50$

The number of headphones sold = $1/2$ of $100 = 50$

So, the total revenue generated from the sale of all earphones = $(50 * 70) = \text{Rs. } 3500$

And, the total revenue generated from the sale of all headphones = $(50 * 150) = \text{Rs. } 7500$

Earphone

Shop	The total number of earphones sold	The total revenue generated from the sale of all earphones (in Rs.)
A	250	25000
B	100	12000
C	200	24000
D	50	3500
Total	600	64500

Headphone

Shop	The total number of headphones sold	The total revenue generated from the sale of all headphones (in Rs.)
A	150	45000
B	100	16000
C	100	20000
D	50	7500
Total	400	88500

Answer: E

Shop B

Let the number of earphones unsold be L.

So, the number of headphones unsold = $(L - 30)$

The number of earphones sold = 150

The number of headphones sold = 100

The total number of earphones manufactured = $(150 + L)$

The total number of headphones manufactured = $100 + (L - 30) = (70 + L)$

Now,

$$(150 + L) / (70 + L) = 5/3$$

$$450 + 3L = 350 + 5L$$

$$2L = 100$$

$$L = 50$$

Now, the total number of earphones manufactured = $(150 + L) = (150 + 50) = 200$

And, the total number of headphones manufactured = $(70 + L) = (70 + 50) = 120$

So, the total number of items manufactured (earphones + headphones) = $(200 + 120) = 320$

@QuizByMickey

The total number of items (earphones + headphones) sold = $(150 + 100) = 250$

Therefore, the required percentage = $250/320 \times 100 = 78.125 \approx 78\%$.

Hence, the correct answer is option E.

3. Questions

Let the number of earphones sold by shop A and B be $5m$ and $3m$ respectively.

Also, the ratio of the number of headphones sold by shop A and B = $150:100 = 3:2$. Let it be $3n$ and $2n$ respectively.

So, the total revenue generated from the sale of all earphones by shop A = $(5m \times 100) = \text{Rs. } 500m$

And, the total revenue generated from the sale of all headphones by shop A = $(3n \times 300) = \text{Rs. } 900n$

Also, the total revenue generated from the sale of all earphones by shop B = $(3m \times 80) = \text{Rs. } 240m$

And, the total revenue generated from the sale of all headphones by shop B = $(2n \times 160) = \text{Rs. } 320n$

Now,

$$500m + 900n = 70000$$

$$5m + 9n = 700 \dots \dots \dots (I)$$

And,

$$240m + 320n = 28000$$

$$3m + 4n = 350 \dots \dots \dots (II)$$

By solving equation (I) and (II), we get,

$$15m + 27n = 2100$$

$$15m + 20n = 1750$$

$$7n = 350$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$5m + 9n = 700$$

$$5m + (9 \times 50) = 700$$

$$5m = 700 - 450$$

$$5m = 250$$

$$m = 50$$

Now, the total number of items sold (earphones + headphones) by shop A = $(5m + 3n) = (5 \times 50 + 3 \times 50) = (250 + 150) = 400$

From the pie chart given above, we have,

The total value of the pie chart = 360°

@QuizByMickey

Now, let the total number of items sold (earphones + headphones) by all four shops together be $360k$.

So, the total number of items sold (earphones + headphones) by shop A = 144° of $360k = 144/360 * 360k = 144k$

Now,

$$144k = 400$$

$$k = (25/9)$$

Therefore, the total number of items sold (earphones + headphones) by all four shops together = $360k = 360 * 25/9 = 1000$.

Shop A

The total number of items sold = 144° of $1000 = 144/360 * 1000 = 400$

The number of earphones sold = $5m = (5 * 50) = 250$

The number of headphones sold = $3n = (3 * 50) = 150$

The total revenue generated from the sale of all earphones = $(250 * 100) = \text{Rs. } 25000$

The total revenue generated from the sale of all headphones = $(150 * 300) = \text{Rs. } 45000$

Shop B

The total number of items sold = 72° of $1000 = 72/360 * 1000 = 200$

The number of earphones sold = $3m = (3 * 50) = 150$

The number of headphones sold = $2n = (2 * 50) = 100$

The total revenue generated from the sale of all earphones = $(150 * 80) = \text{Rs. } 12000$

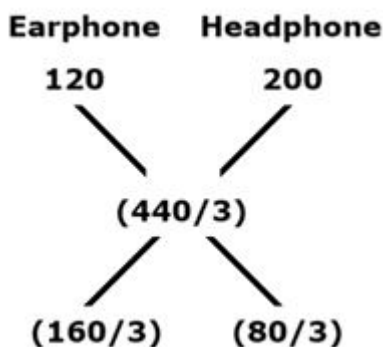
The total revenue generated from the sale of all headphones = $(100 * 160) = \text{Rs. } 16000$

Shop C

The total number of items sold = 108° of $1000 = 108/360 * 1000 = 300$

So, the average price of each item (earphones + headphones) = $(44000/300) = \text{Rs. } (440/3)$

Now, using allegation method, we get,



@QuizByMickey

So, the ratio of the number of earphones and headphones sold = $(160/3) : (80/3) = 160:80 = 2:1$.

The number of earphones sold = $2/3$ of $300 = 200$

The number of headphones sold = $1/3$ of 300 = 100

So, the total revenue generated from the sale of all earphones = $(200 * 120) = \text{Rs. } 24000$

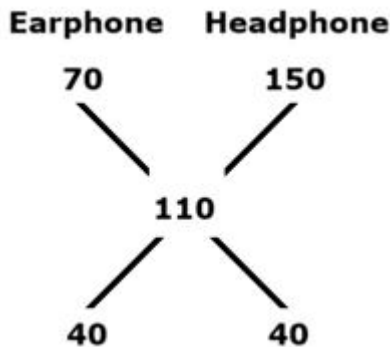
And, the total revenue generated from the sale of all headphones = $(100 * 200) = \text{Rs. } 20000$

Shop D

The total number of items sold = 36° of 1000 = $36/360 * 1000 = 100$

So, the average price of each item (earphones + headphones) = $(11000/100) = \text{Rs. } 110$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $40:40 = 1:1$.

The number of earphones sold = $1/2$ of 100 = 50

The number of headphones sold = $1/2$ of 100 = 50

So, the total revenue generated from the sale of all earphones = $(50 * 70) = \text{Rs. } 3500$

And, the total revenue generated from the sale of all headphones = $(50 * 150) = \text{Rs. } 7500$

Earphone

Shop	The total number of earphones sold	The total revenue generated from the sale of all earphones (in Rs.)
A	250	25000
B	100	12000
C	200	24000
D	50	3500
Total	600	64500

Headphone

Shop	The total number of headphones sold	The total revenue generated from the sale of all headphones (in Rs.)
A	150	45000
B	100	16000
C	100	20000
D	50	7500
Total	400	88500

Answer: A

Shop D

The total number of earphones sold = 50

The total number of headphones sold = 50

The ratio of the number of earphones and headphones sold to females = 2:1. Let it be $2x$ and x respectively.

So, the number of earphones sold to males = $(50 - 2x)$

And, the number of headphones sold to males = $(50 - x)$

Now,

$$(50 - x) - (50 - 2x) = 10$$

$$x = 10$$

Therefore, the number of earphones sold to males by shop D = $(50 - 2x) = (50 - 2 * 10) = (50 - 20) = 30$.

Hence, the correct answer is option A.

4. Questions

Let the number of earphones sold by shop A and B be $5m$ and $3m$ respectively.

Also, the ratio of the number of headphones sold by shop A and B = $150:100 = 3:2$. Let it be $3n$ and $2n$ respectively.

So, the total revenue generated from the sale of all earphones by shop A = $(5m * 100) = \text{Rs. } 500m$

And, the total revenue generated from the sale of all headphones by shop A = $(3n * 300) = \text{Rs. } 900n$

Also, the total revenue generated from the sale of all earphones by shop B = $(3m * 80) = \text{Rs. } 240m$

And, the total revenue generated from the sale of all headphones by shop B = $(2n * 160) = \text{Rs. } 320n$

Now,

$$500m + 900n = 70000$$

$$5m + 9n = 700 \dots\dots\dots(I)$$

And,

$$240m + 320n = 28000$$

$$3m + 4n = 350 \dots\dots\dots (II)$$

By solving equation (I) and (II), we get,

$$15m + 27n = 2100$$

$$15m + 20n = 1750$$

$$7n = 350$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$5m + 9n = 700$$

$$5m + (9 * 50) = 700$$

$$5m = 700 - 450$$

$$5m = 250$$

$$m = 50$$

Now, the total number of items sold (earphones + headphones) by shop A = $(5m + 3n) = (5*50 + 3*50) = (250 + 150) = 400$

From the pie chart given above, we have,

The total value of the pie chart = 360°

Now, let the total number of items sold (earphones + headphones) by all four shops together be 360k.

So, the total number of items sold (earphones + headphones) by shop A = 144° of 360k = $144/360 * 360k = 144k$

Now,

$$144k = 400$$

$$k = (25/9)$$

Therefore, the total number of items sold (earphones + headphones) by all four shops together = $360k = 360 * 25/9 = 1000$.

@QuizByMickey

Shop A

The total number of items sold = 144° of 1000 = $144/360 * 1000 = 400$

The number of earphones sold = $5m = (5 * 50) = 250$

The number of headphones sold = $3n = (3 * 50) = 150$

The total revenue generated from the sale of all earphones = $(250 * 100) = \text{Rs. } 25000$

The total revenue generated from the sale of all headphones = $(150 * 300) = \text{Rs. } 45000$

Shop B

The total number of items sold = 72° of $1000 = 72/360 \times 1000 = 200$

The number of earphones sold = $3m = (3 \times 50) = 150$

The number of headphones sold = $2n = (2 \times 50) = 100$

The total revenue generated from the sale of all earphones = $(150 \times 80) = \text{Rs. } 12000$

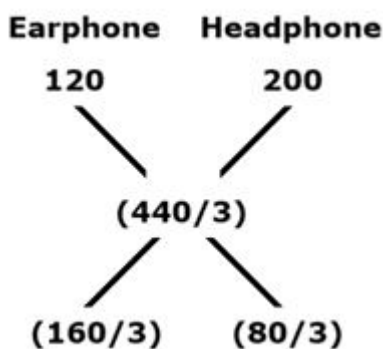
The total revenue generated from the sale of all headphones = $(100 \times 160) = \text{Rs. } 16000$

Shop C

The total number of items sold = 108° of $1000 = 108/360 \times 1000 = 300$

So, the average price of each item (earphones + headphones) = $(44000/300) = \text{Rs. } (440/3)$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $(160/3) : (80/3) = 160:80 = 2:1$.

The number of earphones sold = $2/3$ of $300 = 200$

The number of headphones sold = $1/3$ of $300 = 100$

So, the total revenue generated from the sale of all earphones = $(200 \times 120) = \text{Rs. } 24000$

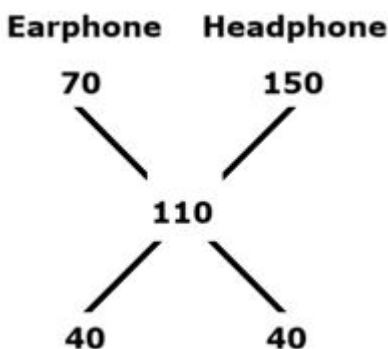
And, the total revenue generated from the sale of all headphones = $(100 \times 200) = \text{Rs. } 20000$

Shop D

The total number of items sold = 36° of $1000 = 36/360 \times 1000 = 100$

So, the average price of each item (earphones + headphones) = $(11000/100) = \text{Rs. } 110$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $40:40 = 1:1$.

@QuizByMickey

The number of earphones sold = $1/2$ of $100 = 50$

The number of headphones sold = $1/2$ of $100 = 50$

So, the total revenue generated from the sale of all earphones = $(50 * 70) = \text{Rs. } 3500$

And, the total revenue generated from the sale of all headphones = $(50 * 150) = \text{Rs. } 7500$

Earphone

Shop	The total number of earphones sold	The total revenue generated from the sale of all earphones (in Rs.)
A	250	25000
B	100	12000
C	200	24000
D	50	3500
Total	600	64500

Headphone

Shop	The total number of headphones sold	The total revenue generated from the sale of all headphones (in Rs.)
A	150	45000
B	100	16000
C	100	20000
D	50	7500
Total	400	88500

Answer: B

Shop A

The overall revenue generated from the sale of all headphones = Rs. 45000

So, the overall manufacturing cost of all headphones = $45000/90 * 100 = \text{Rs. } 50000$

And, the total number of headphones sold = 150

So, the total number of headphones manufactured = $150/3 * 5 = 250$

Therefore, the manufacturing cost of each headphone = $(50000/250) = \text{Rs. } 200$.

So, ? = 200.

Hence, the correct answer is option B.

5. Questions

Let the number of earphones sold by shop A and B be 5m and 3m respectively.

@QuizByMickey

Also, the ratio of the number of headphones sold by shop A and B = $150:100 = 3:2$. Let it be $3n$ and $2n$ respectively.

So, the total revenue generated from the sale of all earphones by shop A = $(5m * 100) = \text{Rs. } 500m$

And, the total revenue generated from the sale of all headphones by shop A = $(3n * 300) = \text{Rs. } 900n$

Also, the total revenue generated from the sale of all earphones by shop B = $(3m * 80) = \text{Rs. } 240m$

And, the total revenue generated from the sale of all headphones by shop B = $(2n * 160) = \text{Rs. } 320n$

Now,

$$500m + 900n = 70000$$

$$5m + 9n = 700 \dots \dots \dots (I)$$

And,

$$240m + 320n = 28000$$

$$3m + 4n = 350 \dots \dots \dots (II)$$

By solving equation (I) and (II), we get,

$$15m + 27n = 2100$$

$$15m + 20n = 1750$$

$$7n = 350$$

$$n = 50$$

Now, put the value of n in equation (I), we get,

$$5m + 9n = 700$$

$$5m + (9 * 50) = 700$$

$$5m = 700 - 450$$

$$5m = 250$$

$$m = 50$$

Now, the total number of items sold (earphones + headphones) by shop A = $(5m + 3n) = (5*50 + 3*50) = (250 + 150) = 400$

@QuizByMickey

From the pie chart given above, we have,

The total value of the pie chart = 360°

Now, let the total number of items sold (earphones + headphones) by all four shops together be $360k$.

So, the total number of items sold (earphones + headphones) by shop A = 144° of $360k = 144/360 * 360k = 144k$

Now,

$$144k = 400$$

$$k = (25/9)$$

Therefore, the total number of items sold (earphones + headphones) by all four shops together = $360k = 360 \times 25/9 = 1000$.

Shop A

The total number of items sold = 144° of $1000 = 144/360 \times 1000 = 400$

The number of earphones sold = $5m = (5 \times 50) = 250$

The number of headphones sold = $3n = (3 \times 50) = 150$

The total revenue generated from the sale of all earphones = $(250 \times 100) = \text{Rs. } 25000$

The total revenue generated from the sale of all headphones = $(150 \times 300) = \text{Rs. } 45000$

Shop B

The total number of items sold = 72° of $1000 = 72/360 \times 1000 = 200$

The number of earphones sold = $3m = (3 \times 50) = 150$

The number of headphones sold = $2n = (2 \times 50) = 100$

The total revenue generated from the sale of all earphones = $(150 \times 80) = \text{Rs. } 12000$

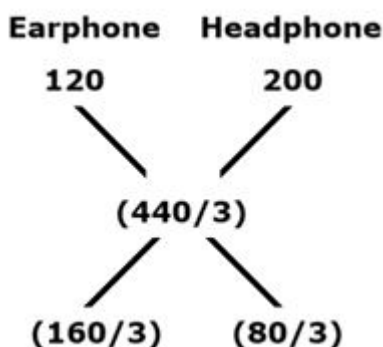
The total revenue generated from the sale of all headphones = $(100 \times 160) = \text{Rs. } 16000$

Shop C

The total number of items sold = 108° of $1000 = 108/360 \times 1000 = 300$

So, the average price of each item (earphones + headphones) = $(44000/300) = \text{Rs. } (440/3)$

Now, using allegation method, we get,



@QuizByMickey

So, the ratio of the number of earphones and headphones sold = $(160/3) : (80/3) = 160:80 = 2:1$.

The number of earphones sold = $2/3$ of $300 = 200$

The number of headphones sold = $1/3$ of $300 = 100$

So, the total revenue generated from the sale of all earphones = $(200 \times 120) = \text{Rs. } 24000$

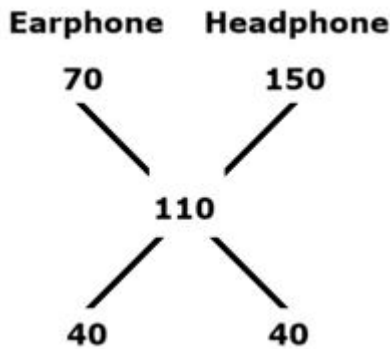
And, the total revenue generated from the sale of all headphones = $(100 \times 200) = \text{Rs. } 20000$

Shop D

The total number of items sold = 36° of $1000 = 36/360 \times 1000 = 100$

So, the average price of each item (earphones + headphones) = $(11000/100) = \text{Rs. } 110$

Now, using allegation method, we get,



So, the ratio of the number of earphones and headphones sold = $40:40 = 1:1$.

The number of earphones sold = $1/2$ of $100 = 50$

The number of headphones sold = $1/2$ of $100 = 50$

So, the total revenue generated from the sale of all earphones = $(50 \times 70) = \text{Rs. } 3500$

And, the total revenue generated from the sale of all headphones = $(50 \times 150) = \text{Rs. } 7500$

Earphone

Shop	The total number of earphones sold	The total revenue generated from the sale of all earphones (in Rs.)
A	250	25000
B	100	12000
C	200	24000
D	50	3500
Total	600	64500

Headphone

Shop	The total number of headphones sold	The total revenue generated from the sale of all headphones (in Rs.)
A	150	45000
B	100	16000
C	100	20000
D	50	7500
Total	400	88500

Answer: C

The number of earphones sold by shop,

$$A = 250$$

$$B = 150$$

$$C = 200$$

$$D = 50$$

So, the average number of earphones sold by all four shops together = $(250 + 150 + 200 + 50)/4 = (650/4) = 162.50$

The number of headphones sold by shop,

$$A = 150$$

$$B = 100$$

$$C = 100$$

$$D = 50$$

So, the average number of headphones sold by all four shops together = $(150 + 100 + 100 + 50)/4 = (400/4) = 100$

Therefore, the required difference = $(162.50 - 100) = 62.50$.

Hence, the correct answer is option C.

6. Questions

Note:

The total number of people who speak languages = the number of people who speak only English + the number of people who speak only Hindi + the number of people who speak both English and Hindi

The total number of people who speak English = the number of people who speak only English + the number of people who speak both English and Hindi

The total number of people who speak Hindi = the number of people who speak only Hindi + the number of people who speak both English and Hindi

Town X

Let the total number of people who speak English be a .

@QuizByMickey

So, the total number of people who speak Hindi = $(a - 120)$

The number of people who speak only English = 200

So, the number of people who speak both English and Hindi = $(a - 200)$

And, the number of people who speak only Hindi = $(a - 200) - 40 = (a - 240)$

Now,

$$(a - 240) + (a - 200) = (a - 120)$$

$$2a - 440 = a - 120$$

$$a = 320$$

Therefore,

The number of people who speak only English = 200

The number of people who speak only Hindi = $(a - 240) = (320 - 240) = 80$

The number of people who speak both English and Hindi = $(a - 200) = (320 - 200) = 120$

The total number of people who speak = $(200 + 80 + 120) = 400$

Town Y

Let the total number of people who speak both English and Hindi be b .

The total number of people who speak English = 350

The total number of people who speak Hindi = $350/140 \times 100 = 250$

So, the number of people who speak only English = $(350 - b)$

And, the number of people who speak only Hindi = $(250 - b)$

Now,

$$(250 - b) / (350 - b) = 3/5$$

$$1250 - 5b = 1050 - 3b$$

$$2b = 200$$

$$b = 100$$

Therefore,

The number of people who speak only English = $(350 - b) = (350 - 100) = 250$

The number of people who speak only Hindi = $(250 - b) = (250 - 100) = 150$

The number of people who speak both English and Hindi = $b = 100$

The total number of people who speak = $(250 + 150 + 100) = 500$

Shop Z

The number of people who speak both English and Hindi = 80% of 100 = 80

Let the number of people who speak only English be c .

So, the number of people who speak only Hindi = $(c - 140)$

The total number of people who speak English = $(c + 80)$

The total number of people who speak Hindi = $(c - 140) + 80 = (c - 60)$

Now,

$$(c + 80) / (c - 60) = 13/6$$

$$6c + 480 = 13c - 780$$

@QuizByMickey

$$7c = 1260$$

$$c = 180$$

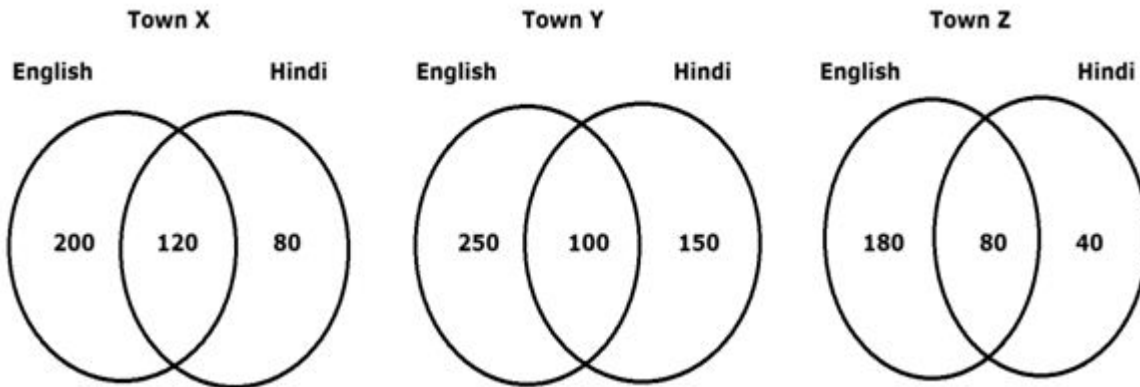
Therefore,

The number of people who speak only English = $c = 180$

The number of people who speak only Hindi = $(c - 140) = (180 - 140) = 40$

The number of people who speak both English and Hindi = 80

The total number of people who speak = $(180 + 40 + 80) = 300$



Answer: B

Now, according to the question,

Town Z

The total number of people who speaks English = $(180 + 80) = 260$

The total number of people who speaks Hindi = $(40 + 80) = 120$

Now, the ratio of the total number of females who speaks English and Hindi = 4:1. Let it be $4n$ and n respectively.

So, the total number of males who speaks English = $(260 - 4n)$

And, the total number of males who speaks Hindi = $(120 - n)$

Now,

$$(260 - 4n) - (120 - n) = 50$$

$$140 - 3n = 50$$

$$3n = 90$$

$$n = 30$$

Therefore, the total number of females who speaks English in town Z = $4n = (4 * 30) = 120$

Hence, the correct answer is option B.

7. Questions

Note:

@QuizByMickey

The total number of people who speak languages = the number of people who speak only English + the number of people who speak only Hindi + the number of people who speak both English and Hindi

The total number of people who speak English = the number of people who speak only English + the number of people who speak both English and Hindi

The total number of people who speak Hindi = the number of people who speak only Hindi + the number of people who speak both English and Hindi

Town X

Let the total number of people who speak English be a .

So, the total number of people who speak Hindi = $(a - 120)$

The number of people who speak only English = 200

So, the number of people who speak both English and Hindi = $(a - 200)$

And, the number of people who speak only Hindi = $(a - 200) - 40 = (a - 240)$

Now,

$$(a - 240) + (a - 200) = (a - 120)$$

$$2a - 440 = a - 120$$

$$a = 320$$

Therefore,

The number of people who speak only English = 200

The number of people who speak only Hindi = $(a - 240) = (320 - 240) = 80$

The number of people who speak both English and Hindi = $(a - 200) = (320 - 200) = 120$

The total number of people who speak = $(200 + 80 + 120) = 400$

Town Y

Let the total number of people who speak both English and Hindi be b .

The total number of people who speak English = 350

The total number of people who speak Hindi = $350/140 \times 100 = 250$

So, the number of people who speak only English = $(350 - b)$

And, the number of people who speak only Hindi = $(250 - b)$

Now,

$$(250 - b) / (350 - b) = 3/5$$

$$1250 - 5b = 1050 - 3b$$

$$2b = 200$$

$$b = 100$$

@QuizByMickey

Therefore,

The number of people who speak only English = $(350 - b) = (350 - 100) = 250$

The number of people who speak only Hindi = $(250 - b) = (250 - 100) = 150$

The number of people who speak both English and Hindi = $b = 100$

The total number of people who speak = $(250 + 150 + 100) = 500$

Shop Z

The number of people who speak both English and Hindi = 80% of 100 = 80

Let the number of people who speak only English be c .

So, the number of people who speak only Hindi = $(c - 140)$

The total number of people who speak English = $(c + 80)$

The total number of people who speak Hindi = $(c - 140) + 80 = (c - 60)$

Now,

$$(c + 80) / (c - 60) = 13/6$$

$$6c + 480 = 13c - 780$$

$$7c = 1260$$

$$c = 180$$

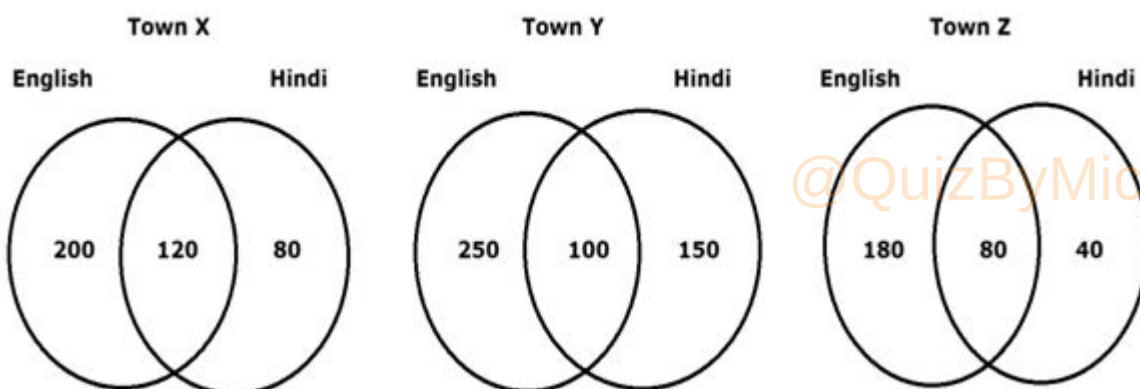
Therefore,

The number of people who speak only English = $c = 180$

The number of people who speak only Hindi = $(c - 140) = (180 - 140) = 40$

The number of people who speak both English and Hindi = 80

The total number of people who speak = $(180 + 40 + 80) = 300$



Answer: C

Town Y

The number of people who speak both English and Hindi = 100

The total number of people who speak languages = 500

Therefore, the required percentage = $100/500 \times 100 = 20\%$.

Hence, the correct answer is option C.

8. Questions

Note:

The total number of people who speak languages = the number of people who speak only English + the number of people who speak only Hindi + the number of people who speak both English and Hindi

The total number of people who speak English = the number of people who speak only English + the number of people who speak both English and Hindi

The total number of people who speak Hindi = the number of people who speak only Hindi + the number of people who speak both English and Hindi

Town X

Let the total number of people who speak English be a .

So, the total number of people who speak Hindi = $(a - 120)$

The number of people who speak only English = 200

So, the number of people who speak both English and Hindi = $(a - 200)$

And, the number of people who speak only Hindi = $(a - 200) - 40 = (a - 240)$

Now,

$$(a - 240) + (a - 200) = (a - 120)$$

$$2a - 440 = a - 120$$

$$a = 320$$

Therefore,

The number of people who speak only English = 200

The number of people who speak only Hindi = $(a - 240) = (320 - 240) = 80$

The number of people who speak both English and Hindi = $(a - 200) = (320 - 200) = 120$

The total number of people who speak = $(200 + 80 + 120) = 400$

Town Y

Let the total number of people who speak both English and Hindi be b .

The total number of people who speak English = 350

The total number of people who speak Hindi = $350/140 \times 100 = 250$

So, the number of people who speak only English = $(350 - b)$

And, the number of people who speak only Hindi = $(250 - b)$

Now,

@QuizByMickey

$$(250 - b) / (350 - b) = 3/5$$

$$1250 - 5b = 1050 - 3b$$

$$2b = 200$$

$$b = 100$$

Therefore,

The number of people who speak only English = $(350 - b) = (350 - 100) = 250$

The number of people who speak only Hindi = $(250 - b) = (250 - 100) = 150$

The number of people who speak both English and Hindi = $b = 100$

The total number of people who speak = $(250 + 150 + 100) = 500$

Shop Z

The number of people who speak both English and Hindi = 80% of 100 = 80

Let the number of people who speak only English be c .

So, the number of people who speak only Hindi = $(c - 140)$

The total number of people who speak English = $(c + 80)$

The total number of people who speak Hindi = $(c - 140) + 80 = (c - 60)$

Now,

$$(c + 80) / (c - 60) = 13/6$$

$$6c + 480 = 13c - 780$$

$$7c = 1260$$

$$c = 180$$

Therefore,

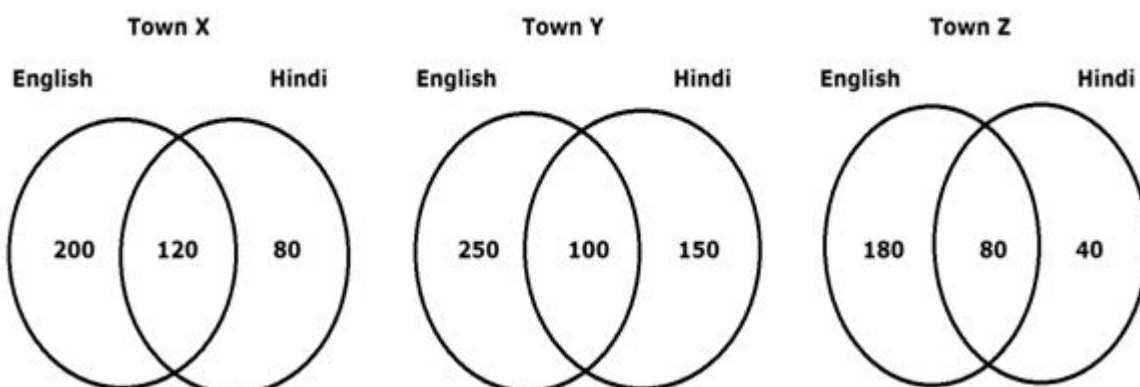
The number of people who speak only English = $c = 180$

The number of people who speak only Hindi = $(c - 140) = (180 - 140) = 40$

The number of people who speak both English and Hindi = 80

The total number of people who speak = $(180 + 40 + 80) = 300$

@QuizByMickey



Answer: D

Town W

The total number of people who speak languages = $(300 * 2) = 600$

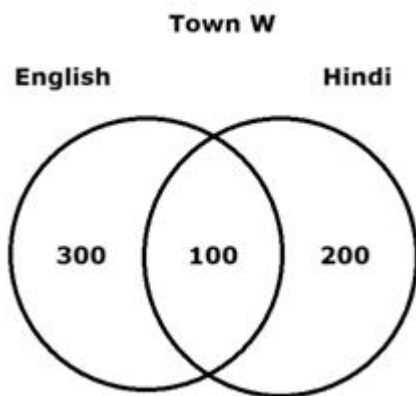
The number of people who speak both English and Hindi = 100

The total number of people who speak Hindi = 300

So, the number of people who speak only Hindi = $(300 - 100) = 200$

Therefore, the number of people who speak only English = $600 - (200 + 100) = (600 - 300) = 300$.

Hence, the correct answer is option D.



9. Questions

Note:

The total number of people who speak languages = the number of people who speak only English + the number of people who speak only Hindi + the number of people who speak both English and Hindi

The total number of people who speak English = the number of people who speak only English + the number of people who speak both English and Hindi

The total number of people who speak Hindi = the number of people who speak only Hindi + the number of people who speak both English and Hindi

Town X

Let the total number of people who speaks English be a .

So, the total number of people who speaks Hindi = $(a - 120)$

The number of people who speak only English = 200

So, the number of people who speak both English and Hindi = $(a - 200)$

And, the number of people who speak only Hindi = $(a - 200) - 40 = (a - 240)$

Now,

$$(a - 240) + (a - 200) = (a - 120)$$

$$2a - 440 = a - 120$$

@QuizByMickey

$$a = 320$$

Therefore,

The number of people who speak only English = 200

The number of people who speak only Hindi = $(a - 240) = (320 - 240) = 80$

The number of people who speak both English and Hindi = $(a - 200) = (320 - 200) = 120$

The total number of people who speak = $(200 + 80 + 120) = 400$

Town Y

Let the total number of people who speak both English and Hindi be b .

The total number of people who speak English = 350

The total number of people who speak Hindi = $350/140 \times 100 = 250$

So, the number of people who speak only English = $(350 - b)$

And, the number of people who speak only Hindi = $(250 - b)$

Now,

$$(250 - b) / (350 - b) = 3/5$$

$$1250 - 5b = 1050 - 3b$$

$$2b = 200$$

$$b = 100$$

Therefore,

The number of people who speak only English = $(350 - b) = (350 - 100) = 250$

The number of people who speak only Hindi = $(250 - b) = (250 - 100) = 150$

The number of people who speak both English and Hindi = $b = 100$

The total number of people who speak = $(250 + 150 + 100) = 500$

Shop Z

The number of people who speak both English and Hindi = 80% of 100 = 80

Let the number of people who speak only English be c .

So, the number of people who speak only Hindi = $(c - 140)$

The total number of people who speak English = $(c + 80)$

The total number of people who speak Hindi = $(c - 140) + 80 = (c - 60)$

Now,

$$(c + 80) / (c - 60) = 13/6$$

$$6c + 480 = 13c - 780$$

@QuizByMickey

$$7c = 1260$$

$$c = 180$$

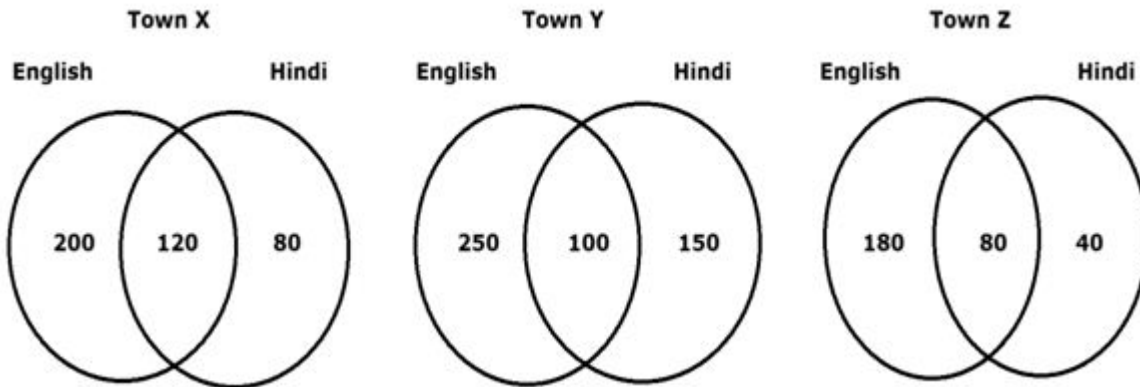
Therefore,

The number of people who speak only English = $c = 180$

The number of people who speak only Hindi = $(c - 140) = (180 - 140) = 40$

The number of people who speak both English and Hindi = 80

The total number of people who speak = $(180 + 40 + 80) = 300$



Answer: E

The number of people who speak only Hindi in town X = 80

The number of people who speak only English in town Z = 180

Therefore, the required difference = $(180 - 80) = 100$.

Hence, the correct answer is option E.

10. Questions

Note:

The total number of people who speak languages = the number of people who speak only English + the number of people who speak only Hindi + the number of people who speak both English and Hindi

The total number of people who speak English = the number of people who speak only English + the number of people who speak both English and Hindi

The total number of people who speak Hindi = the number of people who speak only Hindi + the number of people who speak both English and Hindi

Town X

Let the total number of people who speaks English be a .

So, the total number of people who speaks Hindi = $(a - 120)$

The number of people who speak only English = 200

So, the number of people who speak both English and Hindi = $(a - 200)$

And, the number of people who speak only Hindi = $(a - 200) - 40 = (a - 240)$

Now,

$$(a - 240) + (a - 200) = (a - 120)$$

$$2a - 440 = a - 120$$

$$a = 320$$

Therefore,

The number of people who speak only English = 200

The number of people who speak only Hindi = $(a - 240) = (320 - 240) = 80$

The number of people who speak both English and Hindi = $(a - 200) = (320 - 200) = 120$

The total number of people who speak = $(200 + 80 + 120) = 400$

Town Y

Let the total number of people who speak both English and Hindi be b .

The total number of people who speak English = 350

The total number of people who speak Hindi = $350/140 \times 100 = 250$

So, the number of people who speak only English = $(350 - b)$

And, the number of people who speak only Hindi = $(250 - b)$

Now,

$$(250 - b) / (350 - b) = 3/5$$

$$1250 - 5b = 1050 - 3b$$

$$2b = 200$$

$$b = 100$$

Therefore,

The number of people who speak only English = $(350 - b) = (350 - 100) = 250$

The number of people who speak only Hindi = $(250 - b) = (250 - 100) = 150$

The number of people who speak both English and Hindi = $b = 100$

The total number of people who speak = $(250 + 150 + 100) = 500$

Shop Z

The number of people who speak both English and Hindi = 80% of 100 = 80

Let the number of people who speak only English be c .

So, the number of people who speak only Hindi = $(c - 140)$

The total number of people who speak English = $(c + 80)$

The total number of people who speak Hindi = $(c - 140) + 80 = (c - 60)$

Now,

$$(c + 80) / (c - 60) = 13/6$$

$$6c + 480 = 13c - 780$$

$$7c = 1260$$

$$c = 180$$

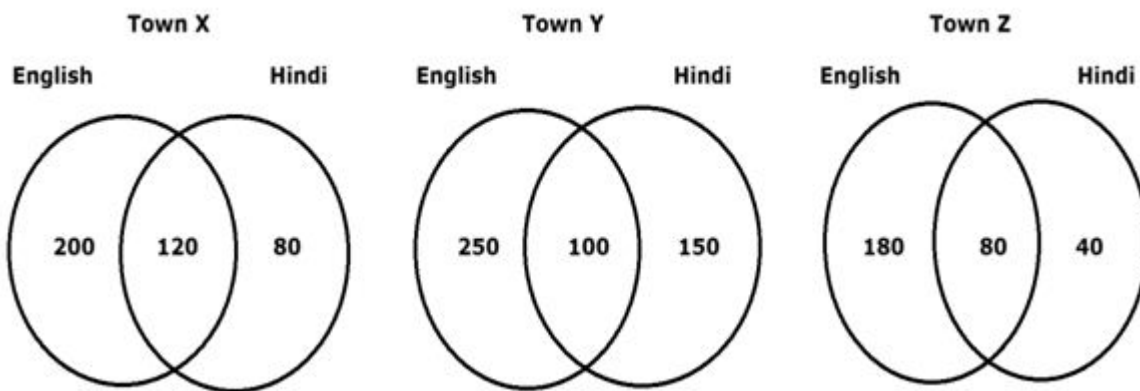
Therefore,

The number of people who speak only English = $c = 180$

The number of people who speak only Hindi = $(c - 140) = (180 - 140) = 40$

The number of people who speak both English and Hindi = 80

The total number of people who speak = $(180 + 40 + 80) = 300$



Answer: A

I: The number of people who speaks both English and Hindi in town X = 120

The number of people who speaks both English and Hindi in town Y = 100

So, the required average = $(120 + 100)/2 = (220/2) = 110$

I = 110.

II: The number of people who speaks only Hindi in town Z = 40

II = 40.

Therefore, $I > II$.

Hence, the correct answer is option A.

11. Questions

Answer: A

Quantity I

Let the total distance covered by the bike be 3T km

@QuizByMickey

So, the distance covered in the first part = $\frac{2}{3}^{\text{rd}}$ of $3T = 2T$ km

And, the distance covered in the second part = $(3T - 2T) = T$ km

The usual speed during the first part of the journey = 40 km/hr

The speed during the second part of the journey = 125% of 40 = 50 km/hr

Now,

$$(2T/40) - (T/50) = 3$$

$$(10T - 4T)/200 = 3$$

$$6T = 600$$

$$T = 100$$

So, the total distance covered by the bike = $3T = (3 * 100) = 300$ km

Quantity I = 300.

Quantity II

The ratio of the speed of the car and bike = 150:100 = 3:2. Let it be $3V$ km/hr and $2V$ km/hr

The time taken to cover the distance by car till stoppage = 10:00 am to 12:00 pm = 2 hours

And, the time taken to cover the distance by the bike from point Y and met the car = 12:00 pm to 2:00 pm = 2 hours

So, the total distance covered by the car = $(3V * 2) = 6V$ km

And, the total distance covered by the bike = $(2V * 2) = 4V$ km

Now,

$$6V - 4V = 48$$

$$2V = 48$$

$$V = 24$$

So, the total distance between point X and Y = $(6V + 4V) = 10V = (10 * 24) = 240$ km.

Quantity II = 240.

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

12. Questions

Answer: A

Quantity I

The total quantity of initial mixture = 200 ml

The initial quantity of milk = 30P ml

@QuizByMickey

The initial quantity of water = $(200 - 30P)$ ml

Resultant Mixture

The quantity of milk = $(30P/2) + P = (15P + P) = 16P$ ml

The quantity of water = $(200 - 30P)/2 + (P + 20) = (100 - 15P) + (P + 20) = (120 - 14P)$ ml

Now,

$$16P / (120 - 14P) = 8/5$$

$$80P = 960 - 112P$$

$$192P = 960$$

$$P = 5$$

$$\text{Quantity I} = P^2 = 5^2 = 25.$$

Quantity II

The total quantity of initial mixture = $40Q$ ml

The initial quantity of milk = $5/8$ of $40Q = 25Q$ ml

The initial quantity of water = $3/8$ of $40Q = 15Q$ ml

Now, the initial quantity of milk taken out = $5/8$ of $40 = 25$ ml

And, the initial quantity of water taken out = $3/8$ of $40 = 15$ ml

So, the quantity of milk in the resultant mixture = $(25Q - 25)$ ml

And, the quantity of water in the resultant mixture = $(15Q - 15) + 40 = (15Q + 25)$ ml

Now,

$$(15Q + 25) - (25Q - 25) = (2Q + 2)$$

$$50 - 10Q = 2Q + 2$$

$$12Q = 48$$

$$Q = 4$$

$$\text{Quantity II} = Q^2 = 4^2 = 16.$$

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

13. Questions

Answer: A

Quantity I

Let the C.P. of the article be Rs. $100n$.

First Case (originally)

@QuizByMickey

The C.P. = Rs. $100n$

The M.P. = 140% of $100n$ = Rs. $140n$

The S.P. = Rs. $(140n - 100)$

Second Case

The C.P. = Rs. $100n$

The M.P. = 120% of $100n$ = Rs. $120n$

The S.P. = Rs. $(120n - 100)$

Now,

$$100n = (120n - 100)$$

$$20n = 100$$

$$n = 5$$

So, the original marked price of the article = $140n = (140 * 5) =$ Rs. 700.

Quantity I = 700.

Quantity II

Let the M.P. of the article be Rs. $100m$.

So, the S.P. = 88% of $100m$ = Rs. $88m$

And, the C.P. = Rs. $(88m - 40)$

Now,

$$(88m - 40) + 100 = 100m$$

$$88m + 60 = 100m$$

$$12m = 60$$

$$m = 5$$

So, the cost price of the article = $(88m - 40) = (88 * 5 - 40) = (440 - 40) =$ Rs. 400.

Quantity II = 400.

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

14. Questions

Answer: E

Quantity I

total runs scored in first five matches = $(5 * 4M) = 20M$

total runs scored in sixth and seventh match together = $(M - 3) + (M - 7) = (2M - 10)$

@QuizByMickey

Now, the total runs scored in all seven matches together = $20M + (2M - 10) = (22M - 10)$

So,

$$(22M - 10)/7 = 75\% \text{ of } 4M$$

$$22M - 10 = 7 * 3M$$

$$22M - 10 = 21M$$

$$M = 10$$

Quantity I = 10.

Quantity II

The total weight of the class initially = $(N * 40) = 40N$

The total weight of 2 students who left = 80 kg

The total weight of 3 new students who joined = $(40 * 3) = 120$ kg

Now, the total weight of the class = $(40N - 80 + 120) = (40N + 40)$ kg

Also, the new number of students in the class = $(N - 2 + 3) = (N + 1)$

So,

$$(40N + 40) / (N + 1) = (N + 10)$$

$$N^2 + 10N + N + 10 = 40N + 40$$

$$N^2 + 11N + 10 = 40N + 40$$

$$N^2 - 29N - 30 = 0$$

$$N^2 - 30N + N - 30 = 0$$

$$N(N - 30) + 1(N - 30) = 0$$

$$(N + 1)(N - 30) = 0$$

$$N = -1 \text{ or } +30$$

So, $N = 30$ (positive integer)

Quantity II = $(N/3) = (30/3) = 10$.

Therefore, Quantity I = Quantity II.

Hence, the correct answer is option E.

15. Questions

Answer: A

Quantity I

The amount invested = Rs. 15000

@QuizByMickey

Let the rate of interest offered in first and second year be $3T\%$ and $2T\%$ respectively.

So, the interest received after first year = $3T\%$ of 15000 = Rs. 450T

And, the interest received after second year = $2T\%$ of 15000 = Rs. 300T

Now,

$$450T - 300T = 1500$$

$$150T = 1500$$

$$T = 10$$

So, the rate of interest offered in second year = $2T\% = (2 * 10)\% = 20\%$.

Quantity I = 20.

Quantity II

The amount invested = Rs. 20000

The rate of interest = $(R + R) + (R * R)/100 = 2R + R^2/100 = (200R + R^2)/100 \%$.

So, the compound interest received = $(200R + R^2)/100 \%$ of 20000 = Rs. $(400R + 2R^2)$

Now,

$$400R + 2R^2 = 420R$$

$$2R^2 = 20R$$

$$R^2 = 10R$$

$$R = 10$$

Quantity II = 10.

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

16. Questions

Answer: B

@QuizByMickey

$$1^{\text{st}} \text{ term} = 20$$

$$2^{\text{nd}} \text{ term} = (20 * 1 + 2) = (20 + 2) = 22$$

$$3^{\text{rd}} \text{ term} = (22 * 2 + 3) = (44 + 3) = 47$$

$$4^{\text{th}} \text{ term} = (47 * 3 + 4) = (141 + 4) = 145 \text{ (144 is wrong)}$$

$$5^{\text{th}} \text{ term} = (145 * 4 + 5) = (580 + 5) = 585$$

$$6^{\text{th}} \text{ term} = (585 * 5 + 6) = (2925 + 6) = 2931$$

So, the wrong term = 144

Hence, the correct answer is option B.

17. Questions

Answer: E

$$1^{\text{st}} \text{ term} = 42$$

$$2^{\text{nd}} \text{ term} = (42 + 11) = 53$$

$$3^{\text{rd}} \text{ term} = (53 - 14) = 39$$

$$4^{\text{th}} \text{ term} = (39 + 17) = 56$$

$$5^{\text{th}} \text{ term} = (56 - 20) = 36 \text{ (35 is wrong)}$$

$$6^{\text{th}} \text{ term} = (36 + 23) = 59$$

So, the wrong term = 35

Hence, the correct answer is option E.

18. Questions

Answer: A

$$1^{\text{st}} \text{ term} = 28$$

$$2^{\text{nd}} \text{ term} = (28/7) = 4$$

$$3^{\text{rd}} \text{ term} = (4 * 14) = 56$$

$$4^{\text{th}} \text{ term} = (56/7) = 8$$

$$5^{\text{th}} \text{ term} = (8 * 14) = 112$$

$$6^{\text{th}} \text{ term} = (112/7) = 16 \text{ (12 is wrong)}$$

So, the wrong term = 12.

Hence, the correct answer is option A.

19. Questions

Answer: C

$$1^{\text{st}} \text{ term} = 11$$

$$2^{\text{nd}} \text{ term} = (11 + 11^2) = (11 + 121) = 132$$

@QuizByMickey

$$3^{\text{rd}} \text{ term} = (132 + 13^2) = (132 + 169) = 301 \text{ (300 is wrong)}$$

$$4^{\text{th}} \text{ term} = (301 + 17^2) = (301 + 289) = 590$$

$$5^{\text{th}} \text{ term} = (590 + 19^2) = (590 + 361) = 951$$

$$6^{\text{th}} \text{ term} = (951 + 23^2) = (951 + 529) = 1480$$

So, the wrong term = 300.

Hence, the correct answer is option C.

20. Questions

Answer: E

$$1^{\text{st}} \text{ term} = 4^2 = 16$$

$$2^{\text{nd}} \text{ term} = 6^3 = 216$$

$$3^{\text{rd}} \text{ term} = 8^2 = 64$$

$$4^{\text{th}} \text{ term} = 100^3 = 1000 \text{ (100 is wrong)}$$

$$5^{\text{th}} \text{ term} = 12^2 = 144$$

$$6^{\text{th}} \text{ term} = 14^3 = 2744$$

So, the wrong term = 100

Hence, the correct answer is option E.

@QuizByMickey